Sahaj Bhadja

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Summary

Developer with 2+ years building **Python data pipelines**, automation tools, analytics workflows, and production systems; delivered tested infrastructure processing 100K+ records, improving performance 30%, and sustaining 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python FastAPI data services** and normalized PostgreSQL schemas for 500+ users, optimizing indexed queries and access patterns to reduce analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Developed **automated data pipelines** processing 5,000+ monthly jobs with incremental loads, reconciliation checks, validation, error handling, and CI/CD deployments, achieving 99%+ reliability across recurring production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **Python Flask processing platform** supporting 10,000+ daily requests, implementing concurrent batch execution and fault-tolerant APIs to process 1,000+ files while maintaining stable performance under peak workloads.
- Owned features across the **full development lifecycle**, translating user requirements into maintainable code, automated validation, persistent results, GitHub Actions deployments, and production troubleshooting for 150+ platform users.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python quantitative-research pipelines** processing 100K+ trajectory records, engineering temporal and spatial features and reproducible evaluation workflows that improved long-horizon planning performance 18% across controlled simulation experiments.
- Analyzed **benchmark metrics and model failure patterns** against three planning architectures, translating quantitative findings into engineering changes that improved closed-loop stability 15% and reduced off-road trajectories during testing.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Troubleshot and optimized **Python API and ETL components** using AWS Glue, S3, SQL, caching, JSON, and Parquet transformations, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Automated **cross-platform builds, tests, and deployments** with Git, Jenkins, PyTest, and Mockito, maintaining 20+ unit/integration suites and consistently reliable releases across Linux, Windows, and macOS production environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Developed **Python Flask APIs for transaction workflows** covering bookings, pricing, payments, refunds, and webhook-driven status updates, maintaining sub-200-millisecond responses across 50,000+ records with real-time Firebase synchronization.
- Built **Databricks and Spark analytics pipelines** over booking and payment events, producing hourly revenue, activity, and demand summaries while reducing manual refund handling 70% through automated reconciliation workflows.

Skills

Programming: Python, C++, SQL, Java, Bash, JavaScript, TypeScript, object-oriented design
Data & Analytics: PostgreSQL, MySQL, MongoDB, Databricks, Spark, AWS Glue, ETL, JSON, CSV, Parquet
Systems: Linux, FastAPI, Flask, REST APIs, distributed systems, concurrency, algorithms, performance optimization
Testing & Delivery: Git, GitHub, PyTest, Mockito, unit/integration testing, CI/CD, Jenkins, GitHub Actions
Cloud: AWS S3, AWS RDS, Docker, Kubernetes, Microsoft Azure, production troubleshooting, Agile/Scrum

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes*
- Built a **financial analytics platform** supporting 50,000+ users and 1M+ transactions, integrating secure bank feeds, real-time synchronization, advanced transaction filtering, MongoDB access, and Kubernetes-based deployment automation.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized Python data-processing API** on AWS ECS/Fargate with CodePipeline, automated validation checks, and CI/CD releases, achieving 87% analytical accuracy and production-grade operational reliability at scale.

CPU Scheduling Simulator Live | *C++, React.js, Algorithms*
- Implemented **six scheduling and optimization algorithms** with interactive controls, Gantt visualization, and dynamic performance metrics, testing 1,000+ process combinations while reducing average turnaround time 25% across benchmark scenarios.